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ADAM MILLARD-BALL

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Gentrification in a Residential Mobility Framework: Social Change, Tenure Change and Chains of Moves in Stockholm

ADAM MILLARD-BALL

Alresford, Colchester, Essex, UK

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ABSTRACT *Studies of the impacts of gentrification have traditionally taken a relatively narrow view, focusing on the gentrifiers who move in and those who they displace. This paper seeks to apply the broader framework of residential mobility, using the vacancy chain concept to examine the knock-on implications across the entire housing market. Taking the case study of Stockholm, it is shown that tenure conversions from rental to co-operative ownership are of limited direct significance for gentrification. However, the indirect effects of this process, through the linked chains of household moves initiated, serve to reinforce other gentrification mechanisms operating within the rental sector, such as allocation mechanisms for rental housing. Therefore, the impacts of gentrification can be significantly underestimated if the wider effects on the housing market are ignored. It is suggested that this residential mobility approach is particularly important if the displacement effects of gentrification and the plight of any losers are to be fully appreciated.*

KEY WORDS: gentrification, displacement, Stockholm

Introduction

Gentrification, briefly understood as the social 'upgrading' of poorer, inner-city neighbourhoods, is a process of social change. For the social composition of an area to change, people need to move. This change can, of course, take place *in situ*, through social mobility (London, 1980), but this route is likely to be slow and of relatively minor importance. Therefore, residential mobility is a key to making gentrification happen.

Residential mobility is indeed implicit in most accounts of gentrification. Neil Smith (1996), for example, describes the 'conquering' of derelict, predominantly black areas of central New York City by white gentrifiers, while Hamnett & Randolph (1988) detail how unscrupulous landlords in London 'persuade' their poorer tenants to leave, allowing their flats to be sold and gentrified. Alternatively, gentrification is sometimes conceptualised in terms of a 'reinvasion' by the affluent of neighbourhoods once abandoned by them (London & Palen, 1984).

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Studies of displacement provide a more explicit link between gentrification and residential mobility, examining the impact on displacees (for example, Atkinson, 2000; Lee & Hodge, 1984; Marcuse, 1986). Rarer in the literature, however, is a more comprehensive approach that acknowledges the 'knock-on' impacts that displaced households may have on the housing market in the rest of the city. In particular, displaced households may increase the pressure on other low-income neighbourhoods, displacing other households out of the city altogether.

This paper sets out a possible approach to analysing these effects of gentrification on the wider housing market, using the example of Stockholm. It enables an understanding of the process at micro-level, through studying the characteristics of the households involved and their motivations for moving; and at an aggregate level, through analysing the implications for social change over Stockholm as a whole. Both the housing market and opportunities for households to move are highly constrained in Stockholm, meaning that the residential mobility approach should be of particular value here. An attempt is made here to adapt the 'vacancy chain' technique to analyse and quantify gentrification-induced displacement in a comprehensive fashion.

It must be emphasised that residential mobility is not an explanation for gentrification in the same way as theories such as the rent and value gaps or the advent of a 'new middle class'. Rather, it highlights that the effects of gentrification are often more far-reaching than generally acknowledged. Existing studies tend to focus on the neighbourhood level, where the effects of the phenomenon are most tangible and politically charged. But gentrification does not merely affect the single property or neighbourhood that is gentrified; the people displaced have to move somewhere, and the consequences can ripple up the 'chain of moves' that is initiated.

After a review of the existing literature on gentrification and displacement, the paper looks at competing explanations for gentrification in Stockholm, and how they might be informed by a residential mobility perspective. It then outlines the methodology, including the vacancy chain concept, and results. The paper concludes by discussing how perceptions of the effects of gentrification are affected by studying the process within this residential mobility framework, particularly in terms of the winners and losers from gentrification.

Gentrification, Residential Mobility and Displacement

By most measures, gentrification is a highly desirable phenomenon. The revitalisation of central cities, improvements in housing quality and the strengthening of the urban taxation base are all reasons why gentrification has often been welcomed (Cassidy, 1980). The drawback is that this revitalisation generally comes at the expense of the removal of the previous inhabitants. While this is glossed over by some as "an unfortunate side effect of middle-class reinvestment in central-city housing" (Lee & Hodge, 1984), displacement is seen by others as the clinching argument against gentrification, one that has so often turned gentrification into a politically contested, hard-fought process (Smith, 1996; Solnit, 2000).

Displacement is implicit in virtually every study of gentrification, as by most definitions gentrification involves an upward transformation in the social status of a neighbourhood. Unless that transformation is due to upward mobility, increased residential density, or current residents moving out for some reason unrelated to gentrification, displacement is the inevitable result.

More explicit work on gentrification and displacement has taken a variety of forms. Many empirical case studies emphasise the nature of the disruption for residents (London & Palen, 1984). Others seek to quantify the problem (Henig, 1984; Lee & Hodge, 1984; LeGates & Hartman, 1986).

A further question for researchers, and one that is most linked with residential mobility, is what happens to the displacees? Smith & LeFaivre (1984) and LeGates & Hartman (1986) conclude that most outmovers cluster around or even within their previous neighbourhoods. Cassidy (1980) describes how gentrification in Capitol Heights, Washington, DC, pushed poor, black households into Prince George's County, Maryland, creating enormous pressures on housing there.

More extensive and conclusive results, however, have been hindered by the lack of data (Engels, 1999; Smith & LeFaivre, 1984). In the words of Atkinson (2000):

> Perhaps the most obvious reason for this research gap it that it is very difficult to track displacees. It is much easier to research the more tangible manifestation of gentrification, given that it is an observable end-state predicated on the removal or voluntary migration of previous residents. (p. 150)

In Engels' (1999) view, this means that countless identical case studies have failed "to examine from an empirical perspective the actual 'process of gentrification' over an extended period of time as opposed to the 'gentrification process' ". While many studies have documented alleged causes and effects, Engels says most "have concerned themselves almost solely with the specific conditions before and after gentrification, not with what happens in between" (p. 1474).

Invisible Displacement

The limited scope of much of this research is amplified when one considers that most studies address only visible displacement. As Marcuse (1986) notes, much gentrification-induced displacement is 'impersonal', caused by increased rents due to 'market trends':

> The tenant is forced to leave, just as much as if the landlord had personally visited him or her and said "Leave, or else!", with a club in his hand. But the force is of the market, not of the club. (p. 162)

Marcuse sets out a typology of four, sometimes overlapping, forms of displacement: (a) direct last-resident displacement, which counts only the last resident in the unit; (b) chain displacement, which also counts previously displaced households occupying the same building; (c) exclusionary displacement, when a household does not even get to move into the building; and (d) displacement pressure, involving households that in effect 'jump before they are pushed'.

These are conceptual definitions, and while distinguishing between them in practice may be difficult, Marcuse argues any analysis of the full impact of displacement must take account of all four. However, he acknowledges that doing so is easier said than done, and his work in New York City is only able to produce a ballpark estimate of 10 000–40 000 displacees.

One of the few other authors to have taken up the challenge of studying exclusionary displacement is Atkinson (2000). He says:

> Displacement also implies a social cost, likely to be in the form of increased housing need, overcrowding in 'hidden households' and homelessness. This is quite apart from the resentment and feelings of disenfranchisement and exclusion that people are likely to feel where they find themselves priced-out of a household; harassed or financially induced to leave a property; or, unable to enter the residential market in an area where they have spent much of their lives or close to a workplace. (p. 163)

Atkinson attempts to overcome the data barrier by combining cross-sectional with longitudinal census data, tracking the flows of gentrifiers and displacees into and out of gentrified areas to move beyond a simple 'before' and 'after' comparison. However, the analysis suffers from the same problems of any study that uses aggregate data. As Atkinson admits, it is impossible to say whether the flows of marginal groups away from the gentrified areas were due to rental increases, landlord harassment or the decision simply to move on.

Engels (1999), in contrast, takes a micro-level approach in his case study of gentrification in Glebe, Sydney. He argues that council rate records, electoral rolls and a household survey allow him to expose the 'inner workings' of gentrification and quantify the extent of displacement, including the displacement of initial 'marginal gentrifiers' by more affluent gentrifiers in later stages of the process. Even this study, however, does not consider the full extent of exclusionary displacement, and the wider effects on the housing market which I seek to explore below.

Filtering and Backward Filtering

An alternative approach to studying the impacts of gentrification, displacement and residential mobility uses the concept of 'backward filtering'. The 'standard' filtering process itself is often used to analyse residential mobility, especially in relation to neighbourhood change. It sees high-income households moving into newly built, higher quality housing, allowing their previous homes to 'filter down' for the benefit of households on the next rung of the housing ladder. This sets a chain of moves into operation, enabling all households in the chain to improve their standard of housing (Hoyt, 1939; Knox, 1995). While filtering theory has been questioned, most criticisms refer to its (mis)use as a basis for housing policy, or its relative importance compared to other processes in the housing market. It is difficult to deny that filtering actually takes place (Clark & Sarapuu, 1984).

The concept of 'backward filtering' involves the rehabilitation of older dwellings, meaning they filter up rather than down (Bysveen & Knutsen, 1987; Clark, 1992; Gale, 1980; Short, 1978). This does not necessarily need to be related to gentrification; backward filtering has been documented in the pre-gentrification context of early 20th century Toronto, for example, due to the upgrading of lower-income workers' self-built housing (Harris, 1990).

Nor does all gentrification have to involve backward filtering. It may occur if gentrifiers 'climb down' the housing ladder to occupy poorer quality housing in the inner city. Yet there is not necessarily a correlation between housing age, location and quality, for example in Stockholm where most inner-city stock was extensively renovated during the 1970s and 1980s. Moreover, gentrifiers are as likely to move into new inner-city housing as to rehabilitate older stock (Smith, 1996). The main problem with the filtering model, and by extension its backward filtering counterpart, is its implicit spatial bias, which assumes that new housing lies in the suburbs and the poorest quality stock in the inner city. However, it remains a useful way to conceptualise the 'chain of moves' involved in gentrification.

Gentrification and the Stockholm Housing Market

While some of Stockholm's inner suburbs are relatively affluent and offer easy access to the urban core, they lack the density, urban vitality and cultural vibrancy of the inner city (defined in line with Utrednings- och Statistikkontoret, 1988). Most suburban development has taken the form of either high-rise apartment buildings or single-family homes, helping to explain why inner-city housing is in high demand.

Extensive gentrification has certainly taken place across inner-city Stockholm. This is borne out by income data, which show a tendency for increasing relative affluence in most inner-city neighbourhoods to be paralleled by increasing relative poverty in the suburbs, particularly those with large concentrations of public housing (Dahlberg, 1995; Regionplane- och trafikkontoret, 2000). It is also demonstrated in a range of small-scale case studies, best summarised in Tonell (1988) and Millard-Ball (2000), which document social and cultural shifts in line with gentrification, as well as income and occupational changes.

The causes of this gentrification, at least on the production side, have often been attributed to tenure conversions from rental to co-operative ownership (Biterman, 1996; Borgegård & Murdie, 1993). Teeland (1988) even goes so far as to state that the extent to which conversions take place is likely to dictate the pace of Swedish gentrification.

In theoretical terms, these tenure conversions can be analysed in terms of the value gap, which refers to the difference in the value of a property under rental compared to co-operative ownership or owner occupation (Clark, 1992; Hamnett & Randolph, 1986; 1988). Rent control means that the relative scarcity of inner-city housing is not reflected in rent levels (Turner, 1996), with willingness to pay often double the actual rent for an inner-city flat (Temaplan AB, 1995). In contrast, co-operative flats are priced at market rate.

Conversion to co-operative ownership closes this value gap and allows the difference to be realised, making such tenure conversions highly profitable. A conversion can be carried out if tenants form an association and purchase the property. The association would purchase the property based on its rental value—the landlord is not permitted to charge the association more than any other purchaser, and no other purchaser would pay more than the rental value. Each household, in contrast, can subsequently sell its individual flat based on its (higher) value under co-operative ownership. (It should be noted that individual tenants may decline to join the association and continue to rent, with the co-operative association as landlord.)

Up to the middle of 1996, 1191 such conversions had been completed since the first in 1970 (Millard-Ball, 2000). The proportion of co-operative flats in the inner city rose from 21 per cent in 1980 to 29 per cent in 1990, reducing public rental to 24 per cent and private rental to 47 per cent (Folk- och bostadsräkning, 1980, 1990). In turn, an increasing proportion of inner-city housing rose beyond the reach of low-income groups, as co-operative flats require a large capital outlay to procure. Conversions from rental to co-operative could therefore be expected to result in greater socio-spatial segregation, with the rich increasingly dominating the inner city.

More recent work, however, has questioned whether these tenure conversions are the most important explanation for gentrification in Stockholm. First, too few properties are converted for the process to have a significant effect on the social composition of the inner city as a whole. Second, the incomes of many residents are already high even before conversion. The latter can be taken to indicate that gentrification within the rental sector has already occurred to some extent (Millard-Ball, 2000).

Indeed, it seems that rental-sector gentrification mechanisms are responsible for most gentrification in Stockholm (Millard-Ball, 2000). Since Swedes tend to be tenure-neutral in their housing preferences, given the strong security granted to tenants (Biterman, 1996; Turner, 1996), gentrification is not the preserve of owner-occupied housing as is often the case elsewhere.

These rental sector gentrification mechanisms include the black market for rental contracts, and the grey market, where under a legal grey area rental contracts can be part-exchanged for a suburban house. Both these mechanisms allow market forces to enter the rental market through the 'commodification' of rental contracts (Lundqvist *et al.*, 1990), enabling gentrifiers to make use of their superior purchasing power. They rely on the fact that rental contracts in inner-city Stockholm are extremely scarce, forcing households to use a wide variety of strategies to obtain them (Figure 1). For example, the municipal housing queue, which serves both the public and private rented sectors, involves a wait of 10 years or more.

A further rental-sector gentrification mechanism is the 'luxury renovation' of rental flats, carried out by landlords to allow them to increase the rent and

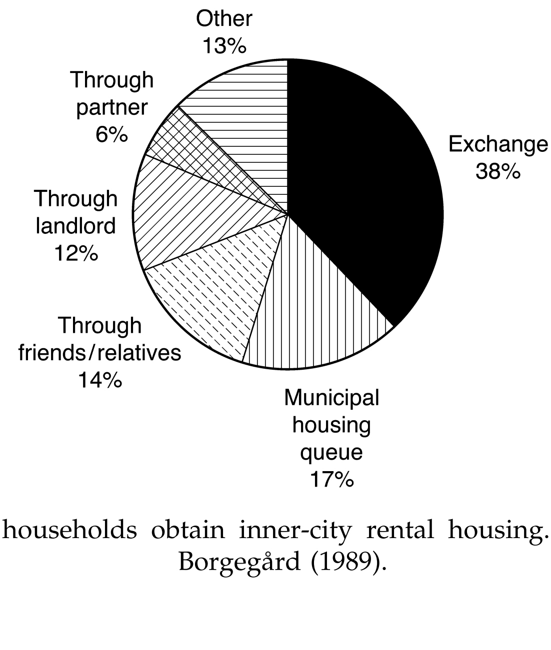

Figure 1. How households obtain inner-city rental housing. *Source:* Siksiö & Borgegård (1989).

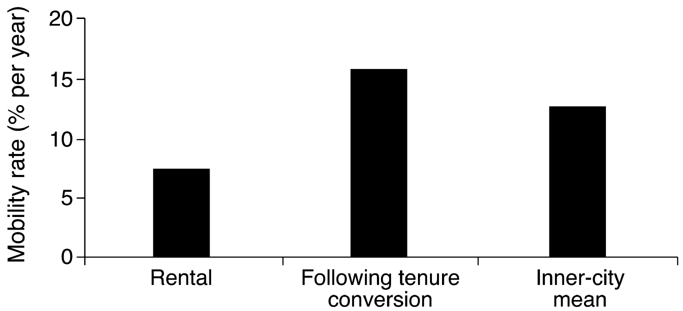

Figure 2. Mobility rates, inner-city Stockholm, 1995. Rental appears to be associated with lock-in problems, and mobility increases substantially for the year following tenure conversion. *Sources:* Olsson (1997); Regionplane- och trafikkontoret (1997).

effectively bypass rent control safeguards. It should be noted that, while sparsely documented in the gentrification literature, these rental sector routes to gentrification are not exclusive to Stockholm (see Engels, 1999, for an Australian example).

The question remains, however, whether tenure conversions would be shown to play a greater role in gentrification if the wider effects of the conversions on the housing market were considered. The analysis above only considers the social changes that take place within the converted property itself. But residential mobility could amplify the gentrification effect, by leading to 'indirect' gentrification and displacement through the chain of moves initiated. For example, a household moving out from the converted property might move in to a flat that would otherwise have been occupied by a poorer household, in turn displacing that less-affluent household to the suburbs. This takes the 'exclusionary displacement' concept (Marcuse, 1986) a stage further, considering exclusionary displacement within the entire housing market.

Consideration of these wider effects is made all the more important by the realisation that tenure conversion also has an impact on residential mobility. Although the reasons were not identified, Olsson (1997) found that household turnover doubles from 7–8 per cent to 16 per cent in the year following conversion (Figure 2). This may partly be due to the relief of lock-in problems, whereby renters are reluctant to move due to the difficulties in obtaining a rental contract. Thus, mobility is lower under private sector rental than other tenures, and residential mobility rates in inner-city Stockholm are extremely low (Sandström, 1991; Siksiö & Borgegård, 1989).

There are two key questions that this paper hopes to answer. First, how does tenure conversion from rental to co-operative ownership affect residential mobility? Is it simply relief of lock-in problems that is behind the increase in mobility? Or is it related to the conversion process in some other way? Second, what is the significance of this mobility for gentrification?

The Vacancy Chain Technique

Gentrification is a form of neighbourhood change. It is, therefore, logical to look to research on other forms of neighbourhood change such as filtering for

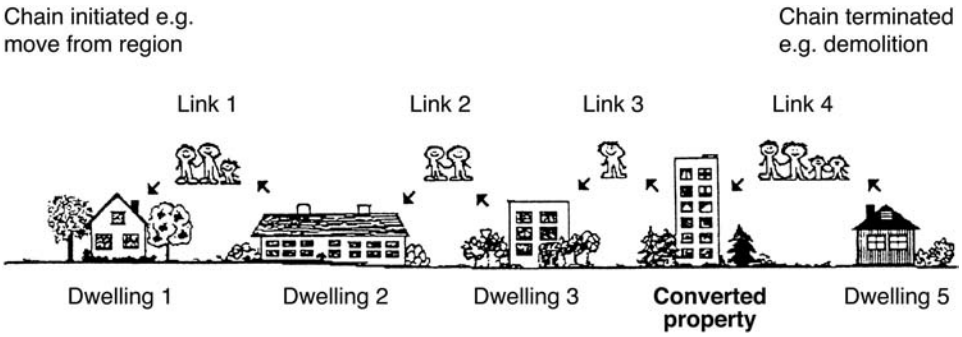

Figure 3. The vacancy chain. *Source:* Adapted from Emmi & Magnusson (1992).

Table 1. Initiation and termination of vacancy chains. Adapted from Magnusson (1994); Clark & Sarapuu (1984)

Initiation: new housing or household disappears	Termination: new household or housing disappears
New housing construction	Demolition
Subdivision of housing units	Consolidation of housing units
Change of building use to housing	Change of building use from housing
Household merging, e.g. cohabitation	Household formation, e.g. after divorce, leaving parental home
Household moves out of region	Household moves into region
Death of all household members	

methodological tools, particularly since 'backward filtering' is one way to conceptualise gentrification.

The vacancy chain concept, based on the linked sequences of household moves, emerges prominently in the literature as one such tool. The chain starts with an empty dwelling, for example a newly-built house (Dwelling 1 in Figure 3). A household (Link 1 in Figure 3) then moves into this empty dwelling, in turn vacating its existing house or apartment (Dwelling 2). The Link 2 household moves into Dwelling 2, and so on until the chain terminates, for example through demolition of an empty dwelling. Table 1 summarises the ways in which chains are initiated and terminated. Household and housing characteristics can be compared at different points in the chain, enabling analysis of neighbourhood change. For example, higher incomes among Link 3 households compared to Link 4 households might indicate filtering; the reverse might indicate gentrification.

Vacancy chain studies have not gone uncriticised. First, on a technical level, Murie *et al.* (1976, p. 91) bemoan the "American fascination with length of chains" to measure the amount of filtering (and thus social benefit) resulting from housing construction. Instead, they suggest that the content of the chains, i.e. who benefits, is far more important in evaluating the effectiveness of filtering in improving housing conditions for all.

A second critique argues that, as a cross-sectional technique, the method is ill-suited to testing filtering theory, which requires a longitudinal approach (Bysveen & Knutsen, 1987; Clark & Sarapuu, 1984). Filtering assumes that older housing is of poorer quality, and that new housing will be successively occupied by lower-status households, but cross-sectional vacancy chain studies consider only one point in a dwelling's lifespan.

Third, while patterns may emerge on the aggregate level, this may not reflect micro-level causal relationships. The characteristics of the fifth vacancy, for example, are not directly caused by those of the first, even though statistically there may be a link (Clark, 1984). Indeed, recent modelling (Emmi & Magnusson, 1988; Magnusson, 1994) implicitly assumes that individual links are independent.

The first technical point is easily overcome by concentrating on chain content as well as length, but do the remaining concerns seriously affect the validity of the technique? Regarding the longitudinal aspect, gentrification theory, unlike filtering, does not rest on longitudinal assumptions of the housing life cycle. Although life cycle assumptions are prevalent in some accounts, particularly those drawing on the rent gap (Clark & Gullberg, 1991; Smith, 1979), they are not a central tenet of gentrification theory. Causality may present greater difficulties, but the problem does not arise for the two links in the vacancy chain that are most important in this context, inmovers to and outmovers from the converted property. While they may not be directly dependent on each other, both links are related to the characteristics of the vacancy in the converted property. Moreover, results from the rest of the chain can still be interesting as an aggregate picture of what does actually happen following tenure conversion, even if a causal relationship is strictly lacking, as long as mechanistic interpretations are avoided (Clark & Sarapuu, 1984).

Data Collection

The three links around the tenure-converted property (Figure 3) are the most significant, as it is here that any changes are most likely to be causally linked to the gentrification process. To recreate these, data collected from various official registers is used. For the remainder of the chains, extrapolations come from secondary data; the absence of causality means that tenure conversion will have little effect on these 'outer reaches' of the vacancy chain. Moreover, with these remaining links, this study is more interested in the general operation of the housing market than with the characteristics of the households involved. The practicalities of this should become clear later; the calculation method is outlined below.

In effect, a two-part analysis of residential mobility resulting from tenure conversion is offered. The first part, at a disaggregate level, focuses on the individual households themselves, why they move and the consequences of this for gentrification. The second, complementary, stage involves broadening out this analysis to an aggregate level, combining the data with secondary sources to examine the likely consequences of the micro-level changes for the socio-spatial residential structure of Stockholm.

Income is used as the prime measure of social status. Although there are problems with measuring such complex concepts as gentrification with a single variable, in this context it is justifiable. Financial barriers largely exclude the poor from co-operatives; income will therefore presumably be the most sensitive measure when studying tenure change. Income indicators have also been successfully used elsewhere (for example, Schaffer & Smith, 1986). While measures such as education might be useful in shedding light on the more cultural aspects of gentrification, the focus here, displacement, is primarily on the economic impacts of tenure change.

The sample comprised nine properties, consisting of 241 flats, drawn from all 29 properties converted from rental to co-operative in inner-city Stockholm between July 1995 and June 1996 and selected primarily on grounds of data availability (see Millard-Ball, 2000, for more details of data collection). Only moves taking place within one year of conversion were considered. This ensures that this period is the same for all properties, regardless of date of conversion, and makes it more probable that the move was related to the conversion process.

Lists of the current residents of each property, obtained from the national registration authority, were compared with details of those resident at the time of conversion, obtained from the economic plans filed by the co-operative association. The date of moving was obtained from the authority, to eliminate those who had moved more than one year after conversion. Names in the property's stairwell were checked to link individual inmovers with individual flats.

In this way, 59 flats were identified that saw a change of resident, and 171 where no one had moved in or out. With 11 flats it was not possible to determine if a move had taken place. The 59 flats saw 71 households move in and 52 move out. Households that moved in together from separate dwellings, or who moved out to separate dwellings, were counted twice, accounting for the larger number of inmoving households compared to flats. Other discrepancies are due to vacant flats, deaths, and lack of data (e.g. where it was possible to determine who moved out but not who moved in, and vice versa).

The previous addresses of inmovers (Dwelling 5 in Figure 3) and the subsequent addresses of outmovers (Dwelling 3) were obtained through a variety of sources, including the national registration authority, census data and the tax authority's database (all of which allow the tracking of individuals), and a reverse telephone directory, the municipal property owner register, the landlord or co-operative association, and the inmoving household. Sources used to 'flesh out' the vacancy chains with data on income, age, tenure, house type and house size included the taxation database, the national registration authority, commercial property directories, the municipal property owner register, the economic plans of co-operative associations, and telephone interviews.

These telephone interviews were conducted with outmovers to establish their reasons for moving. A few short, structured questions to obtain or confirm register data such as tenure were followed by semi-structured questions on the reason for moving and its relationship, if any, to tenure conversion. Few declined to be interviewed, although it was often difficult to contact a household, in particular because of unlisted telephone numbers. The person answering the telephone was taken to represent the household's views, and no systematic gender bias was evident in practice; indeed, most households were single person. The overall response rate was 44 per cent (23 responses). Four additional households can be ascribed a reason for moving with relative certainty (e.g. moving in with partner), even though no interview took place.

Residential Mobility in the Tenure Converted Properties

Fifty-nine of the 241 flats saw a change of resident, corresponding to an annual mobility rate of 24 per cent. This figure is even higher among co-operative association members, compared to those who declined to join the co-operative and continued to rent. Tenure conversion, then, appears to lead to a substantial increase in residential mobility, compared to the 7–8 per cent in inner-city rental flats (Figure 2).

The results also indicate that mobility rates rise more significantly in the case of larger flats. In general, as young people tend to both move more frequently (Regionplane- och trafikkontoret, 1997) and occupy smaller flats, overall mobility should be also greater with these smaller flats. However, this is not borne out by the data here, which indicate similar levels of mobility regardless of flat size, with a slightly lower rate for three-room flats (Figure 4). This suggests that the *increase* in mobility following tenure conversion is particularly pronounced with larger flats (four rooms or more), possibly as the windfall profits incentive is greater.

Why does mobility increase following tenure conversion? As shown in Table 2, many moves result from the so-called life-course events, such as child-raising and moving in with domestic partners, that other studies indicate is the norm (Cadwallader, 1992; Rossi, 1980). There are, however, a number of additional motives specific to tenure conversions—increased housing costs, even though an increase is the exception rather than the norm (Olsson, 1997); realisation of the windfall profits from conversion, whether in cash or through enabling an upgrade to better housing; and threats from the co-operative association. In the latter case, the interviewee claims to have been threatened with punitive increased rent if she continued to rent, rather than take part in the conversion. She moved out soon afterwards due to the unpleasant atmosphere generated.

Much of the increased mobility remains, however, to be accounted for, about which one can only speculate. It is possible that relief of lock-in problems facilitated moves, even if the interviewee was not directly or consciously aware of this. This would mean that they gave, for example, a life-course response, whereas the underlying reason would have been tenure conversion. As discussed above, inner-city rental contracts have a monetary value, but one that is difficult to realise without conversion. Finally, some people may have waited to

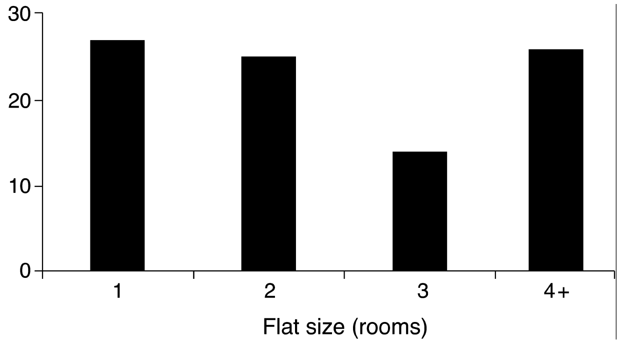

Figure 4. Mobility rate with flat size.

Table 2. Why households moved from the converted properties. Figures only refer to residential mobility within Greater Stockholm, not migration to other parts of the country or abroad (Rossi, 1980)

	Number moving from		
Reason for moving	Co-operative	Rental	Total
Life course	*9*	*1*	*10*
moved in with partner	5	0	5
Separated from partner	2	0	5
more children	1	0	1
"family reasons" (unspecified)	1	0	1
moved to sheltered housing	0	1	1
Housing	*11*	*2*	*13*
too small	5	1	6
desired higher quality	4	1	5
too large and expensive	0	1	1
wanted house, not flat	2	0	2
Conversion process	*6*	*0*	*6*
became too expensive	3	0	3
threats (see text)	1	0	1
to realise conversion profits	2	0	2
poor finances of association	1	0	1
Locational	*3*	*0*	*3*
tired of living in the city	1	0	1
other locational factors	2	0	2
Other	*2*	*0*	*2*
"too many young people"	1	0	1
afraid of the dark, so cannot live alone	1	0	1

Note: Several interviewees gave (unprompted) more than one reason; subtotals may therefore be less than the numbers in individual categories. For the same reason, the subtotals add up to more (34) than the total number of responses (27). Some of the categories are obviously related, increased housing size corresponds to increased household size, for example. However, the imperative was to summarise the interviewee's motive(s) as accurately as possible, using their own words.

move until after conversion in order to share in the windfall, thus creating a backlog of moves.

A Disaggregate Analysis of the Effects of Mobility

What, then, are the direct consequences of this increased mobility, in the narrow terms of socio-economic differences between inmovers and outmovers? As discussed earlier, tenure change theoretically should lead to social 'upgrading', since access to capital is required to secure a co-operative flat. Certainly, inmovers in the sample have mean incomes that are 17 per cent higher than outmovers. Yet the process is not quite so simple. First, median income is

Table 3. Income change with flat size. Swedish kronor/year. Incomes are calculated as the mean of all individual incomes, not household incomes

	Mean income			
Size (rooms)	Inmovers	Stayers	Outmovers	% difference, inmovers: outmovers
1	64 000	116 300	114 500	− 44%
2	183 500	160 200	168 800	+ 9%
3	236 900	215 900	219 800	+ 8%
4 +	307 100	240 700	195 500	+ 57%
Mean	227 400	199 000	187 400	+ 21%

slightly lower among inmovers, indicating that their higher mean incomes result from a small proportion with very high incomes (i.e. a positive skew). Second, on the household level, income is as likely to fall as increase, and numbers of both high- and low-income households increase; there is no automatic pattern of social 'upgrading'. One possibility is that young, single professionals and similar groups are moving in and spending a higher proportion of their income on housing costs, representing gentrification in cultural rather than strict economic terms. However, there are no data either to support or reject this hypothesis.

It seems that gentrification processes operating within the rental sector narrow the income differential between rental and co-operative, reducing the effect of tenure conversion (Millard-Ball, 2000). There are also complementary explanations, such as flat size. With small flats (1–2 rooms), inmovers tend to have lower incomes than outmovers (Table 3). With larger flats, the opposite is true. In other words, the gentrification occurring in larger flats is being partially masked by a reverse process taking place in smaller ones.

Extending the analysis along the chain, Dwelling 3 should theoretically experience a 'standard' filtering process, with income decreasing between Links 2 and 3. Moreover, the Link 3 household can use profits from the tenure conversion to purchase more expensive housing than its income would otherwise allow, housing, presumably, previously occupied by a higher-income household. Indeed, mean income falls by more than 10 per cent between Links 2 and 3. But the picture is again different on the individual household level. Median income falls, income falls in more individual cases than it rises, and the proportions of high, medium and low-income households change little. The highest mean incomes are found among those moving to other co-operative flats in the inner city, excluding those moving to suburban rental for whom the sample is extremely small ($n = 3$).

The Spatial Pattern of Residential Mobility

The importance of the social changes within the converted properties for inner-city gentrification hinges on where people move from and to. If all movements take place within the inner city, the aggregate effect on inner-city gentrification will be nil. This is not to deny the fact that gentrification may also take place on the *property* level within the inner city. However, such a process is of more limited interest from a policy perspective in Stockholm, where the concern is increasing social segregation between the inner city and the suburbs. Some interaction with the suburban housing market is necessary to engender social change in the inner city as a whole.

In many contexts, inner-city gentrification generally results from gentrifiers moving to specific neighbourhoods from elsewhere in the inner city, rather than a 'back-to-the-city' movement (Smith, 1979, 1996). However, gentrification in Stockholm is not clearly neighbourhood-specific, and appears to be a more uniform process affecting most of the inner city (Millard-Ball, 2000), caused perhaps by the scarcity of inner-city housing limiting the choice of neighbourhood by would-be gentrifiers.

In practical terms, this means that the inner-city gentrification process will be governed to a great extent by the proportion of people moving to the inner city from the suburbs. Indeed, it appears that the inmovers with the highest incomes are those who move from other co-operative flats within the inner city (Table 4), indicating that social 'upgrading' within the converted property largely involves people moving from other inner-city flats, and has little significance in terms of wider inner-city gentrification.

Of those moving in to the newly converted properties, more than half come from inner-city Stockholm, with only around a quarter from the suburban municipalities and none from the (generally poorer) southerly municipalities (Table 5). For comparison, 51 per cent of all inmovers to the 'core' housing market (roughly equivalent to inner-city Stockholm) come from suburban municipalities (Regionplane- och trafikkontoret, 1997).

Particularly notable is the large proportion (more than one-third) moving from rented accommodation. Since, generally speaking, renters are less affluent, it is surprising that so many could afford an inner-city co-operative flat. However, it seems that most of these households moved to smaller flats within the converted properties, again highlighting the importance of flat size. With 1 and 2 room flats, nearly two-thirds had previously lived in rental; for larger flats, the proportion is just over a quarter. The pattern becomes even clearer when moves involving two households moving together and moves from the parental home are disregarded, i.e. those moves where previous tenure is less meaningful.

As Table 6 shows, a high proportion of outmovers from the converted properties remain within the inner city (two-thirds) and move to rental flats (nearly half). The former figure differs little from the aggregate picture, where 63 per cent of movers from the 'core' housing market stay within the core (Regionplane- och trafikkontoret, 1997). It is noteworthy, however, that so many move to inner-city rental flats, the type of housing that is most difficult to obtain. It also runs counter to the 'standard' housing career, which sees young people in rented accommodation move to owner-occupation (or co-operatives) as they get older and more affluent. This may be a reflection of the tenure-neutral housing preferences of many Swedes. In any case, the effect on housing availability is clear. The proportion of inner-city rental flats is already reduced through conversions to co-operative ownership; it seems that this proportion is being further whittled away, through the vacancies being taken by outmovers from the tenure-converted properties.

Table 4. Income change with inmovers' previous housing

	Inmovers moving from …						
	Inner city		Rest of Stockholm region				
After conversion, income …	Rental	Co-operative/ owner occupied	Rental	Co-operative/ owner occupied	Outside region	No data	Total
Increases	6	15	0	3	3	0	27
Decreases	5	3	3	7	5	0	23
No data	7	3	3	1	1	6	15
Total	18	21	6	11	9	6	71
Mean income of inmovers	167 500	274 500	215 400	256 400	193 300	—	232 100

Note: In contrast to Table 3, incomes are not the mean of all individuals. Instead, the mean income of all adult earners is calculated for each household, and these household means are then averaged to obtain mean income. The effect is to reduce the significance of households with more than one adult earner, although in practice it makes little difference to the results.

Table 5. Origins of inmovers

Moved in to the converted property from …	Same parish or property*	Elsewhere in inner city	Elsewhere in municipality	Elsewhere in region	Outside region	No data	Total
Rental (flat)	3	15	1	5	0	0	24
Co-operative (flat)	9	11	2	0	0	0	22
Co-operative or owner occupation (house)	0	1	3	6	0	0	10
No data	0	1	0	1	9	4	15
Total	12	28	6	12	9	4	71

Note: *Some households moved between flats in the same property.

Table 6. Destinations of outmovers

Moved from the converted property to …	Same parish	Elsewhere in inner city	Elsewhere in municipality	Elsewhere in region	Outside region	No data	Total
Rental	4	12	1	2	0	0	19
Co-operative/owner occupation	8	6	3	6	0	0	23
No data	0	1	0	1	3	5	10
Total	12	19	4	9	3	5	52

Analysis of the Complete Vacancy Chain

The paper will now take a broader perspective, looking at how conversion affects the ability of different groups to obtain housing in inner-city Stockholm, the indirect effects of tenure conversion, through examining the entire vacancy chain.

Residential mobility depends largely on housing opportunities. Each time a vacancy arises, several households have the opportunity to move in. This can lead gradually to neighbourhood change, for example gentrification. By comparing the full vacancy chain with a 'control' chain, involving a hypothetical identical property that remains under rental rather than being converted to co-operative ownership, conclusions can be drawn as to the change in the number and type of housing opportunities that can be specifically attributed to tenure conversion.

For purposes of analysis, the chains are divided into two parts, vacancies created, through people moving into the converted property (or, for the control chain, the unconverted property), and vacancies absorbed, through households moving into them from the converted property. These absorbed vacancies have arisen through other process within the housing market; households moving into them from the converted property will thus diminish the total number of available vacancies. Created vacancies, on the other hand, represent 'extra' vacancies arising from moves into the converted property. The net effect on housing opportunities will therefore be:

vacancies created − vacancies absorbed

It is vital to appreciate that the figures do not refer to the number of vacancies that actually exist at any one moment, but to the total number of vacancies that *have existed*. Even though most are subsequently absorbed, each represented a potential housing opportunity.

Different types of vacancies (tenure, size, location, etc.) constitute different discrete, homogeneous housing submarkets. Since all housing within a particular submarket is, in theory, identical as far as the household is concerned, there is a constant, defined probability of moving into a specified submarket from a different submarket.

A simple division into three housing submarkets is used here:

- Inner-city rental
- Inner-city co-operative
- Rest of Greater Stockholm, all tenures

While this categorisation is too broad to meet the requirements of homogeneity (Magnusson, 1994), a greater number of categories would lead to problems of data availability, and make the analysis unwieldy.

To calculate the number of vacancies created and absorbed, the data from this study is used to count the number and type of vacancies arising within the converted property and the links either side of it in the vacancy chain. For all other links, the following data are used:

- Estimates for the probability of a chain terminating at a particular link (0.345) and mean chain lengths (1.4 for rental, 1.9 for co-operative), based on data in Clark & Sarapuu (1984). While the data are old, unfortunately no more recent figures are available.

Table 7. Mobility between inner city and suburbs.

	% inmovers from inner city	% inmovers from suburbs
Suburban vacancies	4.3%	95.7%
Inner-city vacancies	48.8%	51.2%

Note: Inmovers from outside Stockholm region are excluded. Calculated from data in Regionplane- och trafikkontoret (1997).

Table 8. Chain lengths resulting from different tenures.

Initial vacancy	Mean chain length	% rental	% co-operative
New rental flats	1.4	97.1%	2.9%
New co-operative flats	1.9	59.6%	40.4%
All new housing	1.9		

Note: Figures exclude owner occupation, which is negligible in the inner city. Chain lengths exclude the initial vacancy. Calculated from data in Clark & Sarapuu (1984).

- The probability of moving between the inner city and suburbs (Table 7), and between different tenures (Table 8).

The number and type of vacancies arising from the initial vacancy in a particular submarket can then be estimated. For example, an initial inner-city rental vacancy will produce 0.663 additional inner-city rental vacancies, calculated as chain length (1.4), multiplied by the probability that an inmover comes from the inner city (0.488), multiplied by the probability that an inmover comes from the rental sector (0.971).

To calculate the number of vacancies absorbed, an iterative process is used, based on the probabilities of moving between different sub-markets, and the probability of a chain terminating at a given link. The results are shown in Table 9.

For the control chain, a similar process is used, with the average mobility in inner-city rental properties used to calculate expected moves from the 'unconverted' property (Table 10). Finally, the figures can (cautiously) be extrapolated to consider the effect of all conversions in a given year in the whole inner city (Table 11).

The net figures are positive, indicating that more vacancies are created than absorbed, with an order of magnitude greater than 1500 per year. The net number of vacancies is certainly higher than for the control chain, suggesting that, although a certain degree of mobility would have taken place anyway, tenure conversion greatly enhances mobility.

This significant increase in overall mobility is in itself a good thing. Although there may be a desire for lower mobility in some contexts, for example to promote more stable neighbourhoods, the policy concern in Stockholm relates to

Table 9. Net vacancies created with tenure conversion

	Submarket			
	Co-operative: inner city	Rental: inner city	Rest of region	Total
Total vacancies created	69.4	34.9	55.1	159.4
Total vacancies absorbed	26.7	25.1	59.8	111.6
Net vacancies created	42.7	9.8	− 4.7	47.8

Table 10. Net vacancies created without tenure conversion (control chain)

	Submarket			
	Co-operative: inner city	Rental: inner city	Rest of region	Total
Total vacancies created	0.4	30.1	13.0	43.5
Total vacancies absorbed	4.6	9.9	19.9	34.4
Net vacancies created	− 4.3	20.2	− 6.9	9.0

Table 11. Effect of tenure conversion on net vacancies

	Submarket			
	Co-operative: inner city	Rental: inner city	Rest of region	Total
Net vacancies created: with conversion	42.7	9.8	− 4.7	47.8
Net vacancies created: conversion (control)	− 4.3	20.2	− 6.9	9.0
Net effect of tenure conversion	+ 47.0	− 10.4	+ 2.2	+ 38.8
Conversions/year (inner-city Stockholm)*		39.3		
Vacancies/year resulting from conversion	+ 1847.1	− 408.7	+ 86.5	+ 1524.8

Note: * Mean value 1994–96.

the shortage of inner-city housing relative to demand, and the desire to allow people who want to move to do so. Policy makers consider that inner-city Stockholm suffers from too little mobility (Regionplane- och trafikkontoret, 1998). However, mobility is only stimulated within the inner-city co-operative sector, it is actually diminished in the case of inner-city rental flats. This is over and above the direct effect of converting individual flats from rental to co-operative ownership.

Undoubtedly, a large margin of error is present in the calculations here. However, the aim was not to provide precise estimates but to highlight trends,

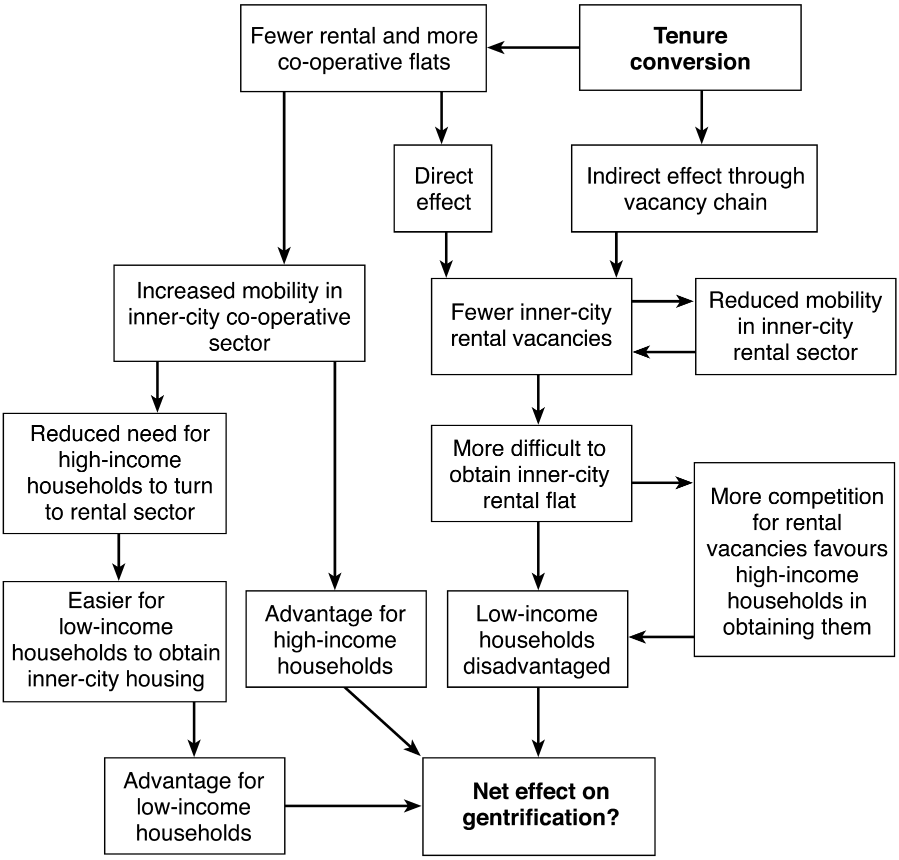

Figure 5. Effect of tenure conversion on the Stockholm housing market.

and these appear to be clear. The direct reason for the reduction in rental vacancies appears to be the large number of households moving out of the converted property to inner-city rental flats, thus absorbing scarce rental vacancies.

The consequences of a reduction in the number of inner-city rental vacancies, summarised in Figure 5, are likely to be threefold.

- As a direct result, the reduced number of vacancies will make it more difficult for low-income households to obtain inner-city housing.
- Competition for the rental vacancies available will stiffen, increasing the significance of the advantages held by higher-income households, such as the right of exchange and the black market. This will indirectly reduce the ability of low-income households to obtain inner-city housing, although this effect will to some extent be offset by an increase in co-operative vacancies, reducing the need for high-income households to turn to the rental sector.
- The mobility of low-income households already living within the inner-city rental sector will probably fall, as they will have fewer opportunities to move elsewhere. This will further reduce the number of rental vacancies.

Conclusion

The effects of tenure conversion are complex and occasionally contradictory. Gentrification certainly occurs through conversion of rental properties to co-operative ownership, even though processes occurring wholly within the rental sector are probably more significant in quantitative terms. Residential mobility increases as both a direct and indirect consequence of tenure conversion, and movers into the tenure-converted properties have, on average, higher incomes than those of outmovers. However, this gentrification is far from a mechanistic process. Even among inmovers, incomes vary greatly; the switch to market housing prices does not automatically entail a social 'upgrading' and displacement of lower-income residents.

Instead, gentrification resulting from tenure change appears to be largely an indirect process, through the change in the number and type of available housing vacancies. The process of tenure conversion leads to a significant increase in residential mobility, at least partly as the windfall profits allow households to upgrade their standard of housing and overcome lock-in problems.

This increased residential mobility has the effect of boosting the numbers of inner-city co-operative vacancies at the expense of an (albeit smaller) reduction in the numbers of inner-city rental vacancies. This further reduces the ability of poorer households to obtain inner-city housing. Even though inner-city rental flats are often taken by the wealthy, this tenure still offers greater chances for low-income households than co-operatives (Turner *et al*., 1993). Moreover, the additional pressure on rental vacancies is likely to reinforce other rental-sector gentrification processes. In particular, the increased scarcity of rental housing will increase the importance of allocation mechanisms that benefit the more affluent, such as the black and grey markets. Therefore, in so far as tenure change is important for gentrification, it probably mainly acts in reinforcing rental-sector gentrification. The converted property itself may not become gentrified, but the changes in patterns of residential mobility contribute to rental-sector gentrification elsewhere within the inner city.

Gentrification through tenure conversion, then, might not seem to be an important policy concern, as it appears to be a win-win situation. Those moving out from the converted properties, who in other contexts would be the displaced losers, both retain security of tenure and receive a handsome windfall profit when they finally move. But there are losers, albeit unseen and unheard—the households who are excluded from the inner-city rental sector, by virtue of the reduced number of vacancies.

Using a residential mobility framework to examine gentrification yields important insights. In the Stockholm context, neglecting the wider housing market would ignore the arena in which much gentrification, albeit indirectly, takes place. It might also lead to the erroneous conclusion that there are, in fact, no losers from tenure conversions. The underlying principle here can almost certainly be translated to other situations; gentrification has a definite impact on housing supply, increasing the quantity available to the affluent and, in most cases, reducing the opportunities available to the poor. Whether it occurs through the rehabilitation of New York brownstones, or the conversion of warehouses in London's Docklands, gentrification has a measurable impact on the wider housing market, and on the ability of different groups to obtain suitable housing at reasonable cost. It influences the prospects of those who cannot in any direct sense be said to be victims of gentrification, not least as they can only be identified in an aggregate sense, rather than individually. While their unseen nature might mean that it is convenient for policy makers, not to mention property owners and developers, to ignore them, the effects of displacement should not be forgotten by researchers.

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Correspondence

Adam Millard-Ball, 8 Wivenhoe Road, Alresford, Colchester CO7 8AD, UK. Email: adam@worldofsebastian.com

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